

PROJECT PROPOSAL

SDF04 – Object-Oriented Programming

Project Title:

Barangay E-Connect Application: A Barangay & resident portal for online document requests,& announcements (offline & Online).

Subject Code	<i>SDF04</i>
Subject Title	<i>Object-Oriented Programming (Python & Database)</i>
Application Type	<i>Desktop / Mobile GUI Application (Non-Console)</i>
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Course / Year / Section	<i>LFCA311N115</i>
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1. Introduction / Background

The background of the problem why I was inspired to do this project is that most of the barangay's is still using the manual style processing of the residents document request, and it usually takes 1 - 5 hours or months of processing documents depends on the volume of residents that is requesting for specific used, but sometimes due to high volume of document request that the staff of barangay can't handle it anymore, and some of the requests that made long ago—where payment has already been settled and the residents are simply waiting for a notification to pick up their documents—are often overlooked.

And when follow-ups are made, staff frequently claim that the requested documents have already been released or have long been ready for pickup; but in reality, however, the requests were forgotten, the documents haven't been printed yet, or they haven't been brought to the barangay captain's office for the necessary signature prior to release. That's why for me this software solution built with Python OOP principles and a database backend is an appropriate response, because this software can make barangay easy to connect to their residents, and update them for what is happening or what program does currently have in their barangay.

2. Project Description

So basically this project with GUI, can make easy for residents to request documents without going to the barangay hall and just wait for an official email notification or app notification that their documents are ready to pickup, and also easy to get an update about what is happening on their barangay, and lastly if there is a problem around them they may report it through the report hub on the app.

3. Objectives

General Objective:

Make every document requested by the residents fast phasing, without going to barangay, and get updates about barangay matters.

Specific Objectives:

1. Design and implement a graphical user interface for [Document Request, updates, and Report Hub].
2. Apply Object-Oriented Programming principles (encapsulation, inheritance, polymorphism, abstraction) in the system's design.
3. Design and integrate a relational normalized PostgreSQL (pgAdmin) database to store Residents Login information and their transaction, and the barangay staff account login details.
4. Implement core CRUD (Create, Read, Update, Delete) functionality through the GUI.
5. Ensure data validation, error handling, and basic security (e.g., login authentication, account creation, requesting) within the application.

4. Scope and Limitations

Scope:

- The application will be a GUI-based [desktop/Mobile app version, real-time multi-user sync] program developed using Python.

- It will include the following core modules: [list modules, e.g., User Authentication, Homepage, Account Creation, Reports, Document Modules, and Transaction].
- Data will be stored and retrieved using a [PostgreSQL] database run by pgAdmin.

Limitations:

- The system will not include [e.g., cloud deployment] unless stated otherwise.
- The system will include only for requesting documents, Announcement, and Reports.

5. System Features / Modules

List and briefly describe each major module or feature of the application.

Module / Feature	Description	Key OOP Class(es) Involved
Login / Authentication (Mobile/Desktop view)	Allows staff of barangay, and residents to securely log in and create accounts online & offline before accessing the Application.	User, Auth
Welcome Page/Announcement (Desktop View)	Allow Barangay staff to create, delete and update an announcement.	Dashboard
Document Request (Desktop View)	Allow Barangay staff to update, and receive the document requested by the residents.	documentRequest
Report Hub (Desktop View)	Allows Barangay staff to add , update or delete their barangay hotlines and contact infos, On this page also can record all the reports and concerns from barangay constituents, and staff can reply to the concerns listed here.	reportHub
Mobile View		
Welcome Page/Announcement (Mobile View)	This page on residents view is a read only, the resident can't do anything Except on reading	Dashboard
Document Request (Mobile View)	The residents can request their documents through this menu, this where also can see updates about their document requested.	documentRequest
Document Request Autolocked Window (Mobile View)	This feature automatically appears when the Target for document requests to process per day has reached its capacity, the document window tab is automatically locked with notices appearing.	autolockedWindow
Email & In-app notifications (Mobile View)	This Feature allow application to send a notification to residents email or app notifications about their requested documents, concerns, or announcement, whether their requested documents is already received, to be signed, or ready for pickup, and the same with the concerns and announcement.	Notifications
Report Hub (Mobile View)	This will allow residents to report their concerns or problem	reportLab

Module / Feature	Description	Key OOP Class(es) Involved
Transaction History (Mobile View)	<i>This feature allows residents to view their previous transactions for requested documents or reporting concerns.</i>	<i>History</i>

6. Tools and Technologies

Programming Language	Python
GUI Framework	<i>[(KivyMD)]</i>
Database	<i>[PostgreSQL]</i>
Database Connector Library	<i>[psycopg2]</i>
IDE / Editor	<i>[VS Code]</i>
Version Control	<i>[Git and GitHub]</i>

7. Project Timeline (Example Estimated)

Phase	Activities	Target Date
Planning & Design	<i>Requirements gathering, ERD, class diagram, UI mockups</i>	<i>[Week 1–3]</i>
Database Setup	<i>Create schema, tables, and sample/test data</i>	<i>[Week 4]</i>
GUI Development	<i>Build application windows/forms and navigation</i>	<i>[Week 5–6]</i>
Backend / OOP Logic	<i>Implement classes, business logic, DB integration</i>	<i>[Week 7]</i>
Testing & Debugging	<i>Unit testing, bug fixing, validation checks</i>	<i>[Week 8]</i>
Documentation & Presentation	<i>Finalize report, prepare demo/presentation</i>	<i>[Week 9]</i>